

Cheeun Hong

Postdoctoral Researcher

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<https://cheeunhong.github.io>



<https://github.com/Cheeun>



[Google Scholar](#)

RESEARCH PROFILE

I develop **efficient AI** methods for reducing the computational cost of deep vision models during training and inference. My work focuses on efficiency optimization, including **quantization, pruning, and adaptive computation**, with applications in image restoration and video generation. I am particularly interested in controllable and content-adaptive compression methods that preserve model quality while enabling practical deployment on resource-constrained devices.

Keywords: *Efficient AI, Model Compression, On-device Vision, Generative Models*

Highlights:

- Developed a series of quantization methods for image restoration, published at CVPR, ECCV, and WACV
- Proposed efficient inference techniques addressing calibration mismatch, adaptive bit allocation, and cost-adaptive quantization in low-level vision models
- Extended efficient AI research toward video generation during a research internship at Meta SuperIntelligence Labs

EDUCATION

Seoul National University — Seoul, Korea

Integrated Ph.D. in Electrical and Computer Engineering | Mar. 2020 - Aug. 2026

Advisor: Prof. Kyoung Mu Lee

Seoul National University — Seoul, Korea

B.S. in Electrical and Computer Engineering | Mar. 2015 - Feb. 2020

University of Applied Sciences and Arts Northwestern Switzerland — Switzerland

Exchange Student in Computer Science | Fall 2017

INDUSTRY & RESEARCH EXPERIENCE

Relational Machine Learning Research Group — CISPA Helmholtz Center, Germany

Postdoctoral Researcher | Oct. 2026 - Present

Supervisor: Rebekka Burkholz

- Efficient LLMs and multi-agent systems through quantization and pruning

Meta SuperIntelligence Labs (MSL) — Meta, Switzerland

Research Scientist Intern | Jun. 2025 - Dec. 2025

Mentor: Albert Pumarola

- Developed a post-training framework for efficient video generation models enabling controllable streaming generation
- Worked on future-guided instructional video generation with a focus on temporal control, streaming inference, and generation efficiency

Machine Intelligence and Pattern Analysis Lab (MIPAL) — Seoul National University, Korea

Student Research Intern | Jun. 2019 - Sep. 2019

Mentor: Prof. Nojun Kwak

- Participated in designing a GAN framework and optimizing it for image generation

Hynix DRAM circuit design team — SK Hynix, Korea

Engineering Intern | Jun. 2018 - Sep. 2018

- Worked on a project on designing efficient verification system and got a Grand Prize Award (1st place) in the internship program among 100+ interns

PUBLICATIONS

[Preprints / Under Review]

Cost-Controllable Quantization for Efficient Image Super-Resolution

Cheeun Hong and Kyoung Mu Lee, Under Review, 2026.

SneakPeek: Future-Guided Instructional Streaming Video Generation

Cheeun Hong, German Barquero, Fadime Sener, Markos Georgopoulos, Edgar Schönfeld, Stefan Popov, Yuming Du, Oscar Mañas, and Albert Pumarola, Under Review, 2026.

[Conferences]

Diversity, Plausibility, and Difficulty: Dynamic Data-Free Quantization

Cheeun Hong*, Sungyong Baik*, Junghun Oh, and Kyoung Mu Lee, In Winter Conference on Applications of Computer Vision (WACV), 2025.

Overcoming Distribution Mismatch in Quantizing Image Super-Resolution Networks

Cheeun Hong and Kyoung Mu Lee, In European Conference on Computer Vision (ECCV), 2024.

AdaBM: On-the-Fly Adaptive Bit Mapping for Image Super-Resolution

Cheeun Hong and Kyoung Mu Lee, In Conference on Computer Vision and Pattern Recognition (CVPR), 2024.

Content-Aware Dynamic Quantization for Image Super-Resolution

Cheeun Hong, Sungyong Baik, Heewon Kim, Seungjun Nah, and Kyoung Mu Lee, In European Conference on Computer Vision (ECCV), 2022. [60+ citations]

Attentive Fine-Grained Structured Sparsity for Image Restoration

Junghun Oh, Heewon Kim, Seungjun Nah, Cheeun Hong, Jonghyun Choi, and Kyoung Mu Lee, In Conference on Computer Vision and Pattern Recognition (**CVPR**), 2022.

DAQ: Channel-Wise Distribution-Aware Quantization for Deep Image Super-Resolution Networks

Cheeun Hong^{*}, Heewon Kim^{*}, Sungyong Baik, Junghun Oh, and Kyoung Mu Lee, In Winter Conference on Applications of Computer Vision (**WACV**), 2022. [80+ citations]

Batch Normalization Tells You Which Filter is Important

Junghun Oh, Heewon Kim, Sungyong Baik, Cheeun Hong, and Kyoung Mu Lee, In Winter Conference on Applications of Computer Vision (**WACV**), 2022.

[Journals]

CoLaNet: Adaptive Context and Latent Information Blending for Face Image Inpainting

JoonKyu Park, Cheeun Hong, Sungyong Baik, and Kyoung Mu Lee, IEEE Signal Processing Letters, 2023.

ACADEMIC SERVICE & ACTIVITIES

- Reviewer: **CVPR** (2022-2026), **ICCV** (2023, 2025), **ECCV** (2022, 2024, 2026), **IEEE TPAMI**, **IEEE TNNLS**
- Technology Transfer: **Fast Deep Super-Resolution Algorithm**, SNU R&DB, 2021

AWARDS & HONORS

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| • Distinguished Dissertation Award (Seoul National University) | 2026 |
| • Youlchon AI Star Scholarship for Core AI (~ \$6000) | 2024 |
| • Best Paper Award at IPIU 2021 (33rd Workshop on Image Processing and Image Understanding) | 2021 |
| • The Grand Prize at Hynix Internship Program | 2018 |

TALKS

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| • AIIS Fall Retreat, SNU (“Content-Aware Dynamic Quantization for Image Super-Resolution”) | 2022 |
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TEACHING EXPERIENCE

Seoul National University

Teaching Assistant in *Recent Trends in Computer Vision*, Spring 2022

Teaching Assistant in *Introduction to Computer Vision*, Spring 2022

REFERENCES

PhD Advisor **Kyoung Mu Lee**
Distinguished Professor
Seoul National University
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<https://cv.snu.ac.kr/index.php/kmlee>

